

DAY7: List Operations & Methods

→ Collection of different datatypes that separated by , and it is represented by []

1.Indexing

positive → 0

```
all_=[12,[1,'python',[1,4]],[78,[6,7]],['java',78]]
```

```
print(all_[1][2])
```

```
data=['python',[1,2,(90,'Details',[67,0])],[78,'student']]
```

```
print(data[1][2][1][2])
```

negative → -1

```
so =[1,2,3,4,'python']
```

```
print(so[-1][-3])
```

2.len()

→ The function is used to find the number of items present inside list

syntax → len(variable_name)

Eg:

```
data=['python',[1,2,(90,'Details',[67,0])],[78,'student']]
```

```
print(len(data))
```

```
data=['python',[1,2,(90,'Details',[67,0])],[78,'student']]
```

```
print(len(data[1][2]))
```

3.Slicing

```
data=[1,2,3,4,5,6,7]
```

```
print(data[1:5])
```

4.concatenation

```
a=[1,2]
b=[3,4]
print(a+b)
```

Methods

1.append()

→ append method will add new items into the list at last index position

syntax → variable_name.append(item)

Eg:

```
go=[1,2,3,4]
print(go)
go.append(5)
print(go)
```

2.extend()

→ extend() will add the items into a list at last index position, but it will give each value as one index inside the list

syntax → variable_name.extend(items)

Eg:

```
go=[1,2]
print(go)
go.extend('python')
print(go)
```

3.pop ()

→ pop () is used to remove items from the list and it will delete based on the index position

syntax → variable_name.pop(index_position)

Eg:

```
m=[1,2,3,4,'python']
m.pop(4)
```

```
print(m)
```

4.Remove()

→ remove() will delete items based on the value given init..

syntax→variable_name(item)

Eg:

```
m=[1,2,3,4,'python']
```

```
m.remove(4)
```

```
print(m)
```